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$$1) (a) F \subseteq P(A) = \forall x (x \in F \rightarrow x \in P(A)) \\ = \forall x (x \in F \rightarrow \forall y (y \in x \rightarrow y \in A))$$

$$(b) A \subseteq \{2n+1 \mid n \in \mathbb{N}\} \\ = \forall x (x \in A \rightarrow x \in \{2n+1 \mid n \in \mathbb{N}\}) \\ = \forall x (x \in A \rightarrow \exists n \in \mathbb{N} (x = 2n+1))$$

$$(c) \{n^2+n+1 \mid n \in \mathbb{N}\} \subseteq \{2n+1 \mid n \in \mathbb{N}\} \\ = \forall x (x \in \{n^2+n+1 \mid n \in \mathbb{N}\} \rightarrow x \in \{2n+1 \mid n \in \mathbb{N}\}) \\ = \forall x (\exists n \in \mathbb{N} (x = n^2+n+1) \rightarrow \exists n \in \mathbb{N} (x = 2n+1))$$

or simply $\forall n \in \mathbb{N} \exists m \in \mathbb{N} (n^2+n+1 = 2m+1)$ (=Translated logically)

$$(d) P(\bigcup_{i \in I} A_i) \not\subseteq \bigcup_{i \in I} P(A_i)$$

$$= \exists x (x \in P(\bigcup_{i \in I} A_i) \wedge x \notin \bigcup_{i \in I} P(A_i)) \\ = \exists x (x \subseteq \bigcup_{i \in I} A_i \wedge \neg \exists i \in I (x \subseteq P(A_i))) \\ = \exists x (\forall y (y \in x \rightarrow y \in \bigcup_{i \in I} A_i) \wedge \forall i \in I (x \not\subseteq P(A_i))) \\ = \exists x (\forall y (y \in x \rightarrow \exists i \in I (y \in A_i)) \wedge \forall i \in I (x \not\subseteq A_i)) \\ = \exists x (\forall y (y \in x \rightarrow \exists i \in I (y \in A_i)) \wedge \forall i \in I \exists y (y \in x \wedge y \notin A_i))$$

$$2) (a) x \in \bigcup F \setminus \bigcup G$$

$$= (x \in \bigcup F \wedge x \notin \bigcup G) \\ = (\exists A \in F (x \in A) \wedge \neg \exists A \in G (x \in A)) \\ = (\exists A \in F (x \in A) \wedge \forall A \in G (x \notin A))$$

$$(b) \{x \in B \mid x \notin C\} \in P(A)$$

$$(x \in B \wedge x \notin C) \in P(A)$$

$$(x \in B \wedge x \notin C) \subseteq A$$

$$\forall y [(x \in B \wedge x \notin C) \rightarrow y \in A]$$

$$(c) x \in \bigcap_{i \in I} (A_i \cup B_i)$$

$$= \forall i \in I (x \in A_i \cup B_i)$$

$$= \forall i \in I (x \in A_i \vee x \in B_i)$$

$$(d) x \in (\bigcap_{i \in I} A_i) \cup (\bigcap_{i \in I} B_i)$$

$$x \in \bigcap_{i \in I} A_i \vee x \in \bigcap_{i \in I} B_i$$

$$\forall i \in I (x \in A_i) \vee \forall i \in I (x \in B_i)$$

$$3) P(\{\emptyset\})? = \{\emptyset, \{\emptyset\}\}$$

$$4) \cap \text{book}$$

$$I = \{2, 3, 4, 5\}$$

$$(a) A_i = \{i, i+1, i-1, 2i\}$$

$$A_2 = \{2, 3, 1, 4\}$$

$$A_3 = \{3, 4, 2, 6\}$$

$$A_4 = \{4, 5, 3, 8\}$$

$$A_5 = \{5, 6, 4, 10\}$$

$$(b) \bigcap_{i \in I} A_i = A_2 \cap A_3 \cap A_4 \cap A_5$$

$$= \{4\}$$

$$\bigcup_{i \in I} A_i = A_2 \cup A_3 \cup A_4 \cup A_5$$

$$= \{1, 2, 3, 4, 5, 6, 8, 10\}$$

$$7. \bigcup_{b \in B} L_b = \{t \mid \exists b \in B (t \in L_b)\} = \{t \in S \mid \exists b(b \in B \wedge L(b, t))\}.$$

But B was defined to be the set of all students who are admired by all students, so $b \in B$ means $b \in S \wedge \forall s \in S A(s, b)$. Inserting this, we get

$$\bigcup_{b \in B} L_b = \{t \in S \mid \exists b(b \in S \wedge \forall s \in S A(s, b) \wedge L(b, t))\} = \text{the set of all students who are liked by some student who is admired by all students.}$$

*The set of all students t such that b where all students like b & b likes t .
i.e. t is taken from set of students admired by all.*

Exercises

- *1. Analyze the logical forms of the following statements. You may use the symbols $\in$, $\notin$, $=$, $\neq$, $\wedge$, $\vee$, $\rightarrow$, $\leftrightarrow$, $\forall$, and $\exists$ in your answers, but not $\subseteq$, $\subset$, $\mathcal{P}$, $\cap$, $\cup$, $\setminus$, $\{\}$, or $\neg$. (Thus, you must write out the definitions of some set theory notation, and you must use equivalences to get rid of any occurrences of $\neg$.)

(a) $\mathcal{F} \subseteq \mathcal{P}(A)$.

(b) $A \subseteq \{2n + 1 \mid n \in \mathbb{N}\}$.

(c) $\{n^2 + n + 1 \mid n \in \mathbb{N}\} \subseteq \{2n + 1 \mid n \in \mathbb{N}\}$.

(d) $\mathcal{P}(\bigcup_{i \in I} A_i) \not\subseteq \bigcup_{i \in I} \mathcal{P}(A_i)$.

2. Analyze the logical forms of the following statements. You may use the symbols $\in$, $\notin$, $=$, $\neq$, $\wedge$, $\vee$, $\rightarrow$, $\leftrightarrow$, $\forall$, and $\exists$ in your answers, but not $\subseteq$, $\subset$, $\mathcal{P}$, $\cap$, $\cup$, $\setminus$, $\{\}$, or $\neg$. (Thus, you must write out the definitions of some set theory notation, and you must use equivalences to get rid of any occurrences of $\neg$.)

(a) $x \in \bigcup \mathcal{F} \setminus \bigcup \mathcal{G}$.

(b) $\{x \in B \mid x \notin C\} \in \mathcal{P}(A)$.

(c) $x \in \bigcap_{i \in I} (A_i \cup B_i)$.

(d) $x \in (\bigcap_{i \in I} A_i) \cup (\bigcap_{i \in I} B_i)$.

3. We've seen that $\mathcal{P}(\emptyset) = \{\emptyset\}$, and $\{\emptyset\} \neq \emptyset$. What is $\mathcal{P}(\{\emptyset\})$? *$\{\emptyset, \{\emptyset\}\}$*

- *4. Suppose $\mathcal{F} = \{\{\text{red, green, blue}\}, \{\text{orange, red, blue}\}, \{\text{purple, red, green, blue}\}\}$. Find $\bigcap \mathcal{F}$ and $\bigcup \mathcal{F}$. *$\bigcap \mathcal{F} = \{\text{red, blue}\}$ $\bigcup \mathcal{F} = \{\text{red, green, blue, orange, purple}\}$*

5. Suppose $\mathcal{F} = \{\{3, 7, 12\}, \{5, 7, 16\}, \{5, 12, 23\}\}$. Find $\bigcap \mathcal{F}$ and $\bigcup \mathcal{F}$. *$\bigcap \mathcal{F} = \{7\}$ $\bigcup \mathcal{F} = \{3, 5, 12, 16, 23\}$*

6. Let $I = \{2, 3, 4, 5\}$, and for each $i \in I$ let $A_i = \{i, i + 1, i - 1, 2i\}$. *$\bigcap_{i \in I} A_i = \{2\}$*

(a) List the elements of all the sets A_i , for $i \in I$.

(b) Find $\bigcap_{i \in I} A_i$ and $\bigcup_{i \in I} A_i$.

7. Let $P = \{\text{Johann Sebastian Bach, Napoleon Bonaparte, Johann Wolfgang von Goethe, David Hume, Wolfgang Amadeus Mozart, Isaac Newton, George Washington}\}$ and let $Y = \{1750, 1751, 1752, \dots, 1759\}$. For each $y \in Y$, let $A_y = \{p \in P \mid \text{the person } p \text{ was alive at some time during the year } y\}$. Find $\bigcup_{y \in Y} A_y$ and $\bigcap_{y \in Y} A_y$.

7) According to birth & death dates of these people

$A_{1750} = \{ \text{Johann Scherstiern, Johann Wolfgang, David Hume, George Washington} \}$

$A_{1751} = \{ \text{Johann Wolfgang, David Hume, } \cancel{\text{David}} \text{ George Washington} \}$

$A_{1752} = \{ \text{Johann Wolfgang, David Hume, George Washington} \}$

$A_{1753} = \{ \quad , \quad , \quad \}$

$A_{1754} = \{ \quad , \quad , \quad \}$

$A_{1755} = \{ \quad , \quad , \quad \}$

$A_{1756} = \{ \quad , \quad , \text{Wolfgang Amadeus, George Wash} \}$

$A_{1757} = \{ \quad , \quad , \quad , \quad \}$

$A_{1758} = \{ \quad , \quad , \quad , \quad \}$

$A_{1759} = \{ \quad , \quad , \quad , \quad \}$

$\cup_{y \in Y} A_y = \{ \text{All except Napoleon Bonaparte \& Isaac Newton} \}$

$\cap_{y \in Y} A_y = \{ \text{Johann Wolfgang, David Hume, George Washington} \}$

8) $I = \{2, 3\}$ $A_i = \{i, 2i\}$ $B_i = \{i, i+1\}$

(a) $A_2 = \{2, 4\}$ $B_2 = \{2, 3\}$

$A_3 = \{3, 6\}$ $B_3 = \{3, 4\}$

(b) $\cap_{i \in I} (A_i \cup B_i) = (A_2 \cup B_2) \cap (A_3 \cup B_3)$

$= \{2, 3, 4\} \cap \{3, 4, 6\}$

$= \{3, 4\}$

$(\cap_{i \in I} A_i) \cup (\cap_{i \in I} B_i) = (A_2 \cap A_3) \cup (B_2 \cap B_3)$

$$= \{\cancel{a}, \cancel{b}, \cancel{c} \} \neq \{a, b, c\}$$

$$= \{3\}$$

These are not same

(c) True 2 " " - The first one says for every index i , $x \in A_i$ or $x \in B_i$. The second says x belongs to every A_i or x belongs to every B_i .

$$9) x \in \bigcup_{i \in I} (A_i \setminus B_i) \quad (1)$$

$$= \exists i \in I (x \in A_i \setminus B_i)$$

$$= \exists i \in I (x \in A_i \wedge x \notin B_i)$$

Note: Existential quantifiers connect distinct over conjunction.

$$x \in (\bigcup_{i \in I} A_i) \setminus (\bigcup_{i \in I} B_i) \quad (2)$$

$$\exists i \in I (x \in A_i) \wedge \neg \exists i \in I (x \in B_i)$$

$$\exists i \in I (x \in A_i) \wedge \forall i \in I (x \notin B_i)$$

$$x \in (\bigcup_{i \in I} A_i) \setminus (\bigcap_{i \in I} B_i) \quad (3)$$

$$x \in \bigcup_{i \in I} A_i \wedge x \notin \bigcap_{i \in I} B_i$$

$$\exists i \in I (x \in A_i) \wedge \neg \forall i \in I (x \in B_i)$$

$$\exists i \in I (x \in A_i) \wedge \exists i \in I (x \notin B_i)$$

Some of them are equivalent

$$(b) \bigcup_{i \in I} (A_i \setminus B_i) = (A_2 \setminus B_2) \cup (A_3 \setminus B_3) \\ = \{4\} \cup \{6\} = \{4, 6\}$$

$$(\bigcup_{i \in I} A_i) \setminus (\bigcup_{i \in I} B_i) = (A_2 \cup A_3) \setminus (B_2 \cup B_3) \\ = \{2, 3, 4, 6\} \setminus \{2, 3, 4\} \\ = \{6\}$$

$$(\bigcup_{i \in I} A_i) \setminus (\bigcap_{i \in I} B_i) = (A_2 \cup A_3) \setminus (B_2 \cap B_3) \\ = \{2, 3, 4, 6\} \setminus \{3\} \\ = \{2, 4, 6\}$$

No they are not equivalent

10) The sets in ex 8

$$\bigcup_{i \in I} (A_i \cap B_i) = (A_2 \cap B_2) \cup (A_3 \cap B_3) \\ = \{2\} \cup \{3\} = \{2, 3\}$$

$$(\bigcup_{i \in I} A_i) \cap (\bigcup_{i \in I} B_i) = \{2, 3, 4, 6\} \cap \{2, 3, 4\}$$

hence they are not equal $\{2, 3, 4\}$

$$(i) \quad x \in P(A \cap B) \\ = x \subseteq A \cap B$$

$$\forall y [y \subseteq x \rightarrow y \subseteq (A \cap B)] \\ \forall y [y \subseteq x \rightarrow (y \subseteq A \wedge y \subseteq B)]$$

$$x \subseteq P(A) \cap P(B)$$

$$x \subseteq P(A) \wedge x \subseteq P(B)$$

$$x \subseteq A \wedge x \subseteq B$$

$$\forall y (y \subseteq x \rightarrow y \subseteq A) \wedge \forall y (y \subseteq x \rightarrow y \subseteq B)$$

~~hence~~

$$= \forall y [(y \in x \rightarrow y \in A) \wedge (y \in x \rightarrow y \in B)]$$

$\therefore$ Universal quantifier distributes over conjunction

$$= \forall y [(y \notin x \vee y \in A) \wedge (y \notin x \vee y \in B)]$$

$$= \forall y [\neg (y \in x) \vee (y \in A \wedge y \in B)]$$

Since these both are equal

$$= \forall y [y \in x \rightarrow (y \in A \wedge y \in B)]$$

12) (Imp) make answer $A = \{1, 2\}$, $B = \{2, 3\}$

$$P(A \cup B) = P(\{1, 2, 3\}) = \{ \emptyset, \{1\}, \{2\}, \{3\}, \{1, 2\}, \{1, 3\}, \{2, 3\}, \{1, 2, 3\} \}$$

$$P(A) \cup P(B) = \{ \emptyset, \{1\}, \{2\}, \{1, 2\} \} \cup \{ \emptyset, \{2\}, \{3\}, \{2, 3\} \}$$

$$= \{ \emptyset, \{1\}, \{2\}, \{3\}, \{1, 2\}, \{2, 3\} \}$$

In logical form

$$P(A \cup B) \text{ is } \forall y [y \in x \rightarrow (y \in A \vee y \in B)]$$

not $P(A) \cup P(B)$ is $\forall y (y \in x \rightarrow y \in A) \vee \forall y (y \in x \rightarrow y \in B)$

The first means, if we take any element of x ($y \in x$) (which is a set itself) (a subset), then that element will belong to A or B .

The second means if we take any x , then its all elements will belong to A or all will belong to B (obviously not both)

13)

$$(a) \bigcup_{i \in I} (A_i \cup B_i) = (\bigcup_{i \in I} A_i) \cup (\bigcup_{i \in I} B_i)$$

$$1- \exists i \in I (x \in A_i \cup B_i) \quad 2- \exists i \in I (x \in A_i) \vee \exists i \in I (x \in B_i)$$

$$\exists i \in I (x \in A_i \vee x \in B_i) = \exists i \in I (x \in A_i \vee x \in B_i)$$

($\therefore$ Existential Quantifier distributes over disjunction.)

$$\rightarrow \exists i \in I (x \in A_i) \vee \exists i \in I (x \in B_i)$$

$$\bigcap_{i \in I} A_i \vee \bigcap_{i \in I} B_i$$

$$(\bigcap_{i \in I} A_i) \cup (\bigcap_{i \in I} B_i)$$

Intuitively from
one side

$$(b) (\bigcap F) \cap (\bigcap G) = \bigcap (F \cap G)$$

$$1- \forall x \in F (x \in A) \wedge \forall x \in G (x \in A)$$

$$\forall x \in F (x \in A)$$

$$2- \forall x \in (F \cap G) (x \in A) \quad \text{R.H.S}$$

$$\forall (x \in F \wedge x \in G) (x \in A)$$

$$\forall x \in F (x \in A) \wedge \forall x \in G (x \in A)$$

$$\bigcap F \cap \bigcap G$$

$$= \bigcap F \cap \bigcap G = \text{L.H.S}$$

$$(c) \bigcap_{i \in I} (A_i \setminus B_i) = (\bigcap_{i \in I} A_i) \setminus (\bigcup_{i \in I} B_i)$$

$$\text{L.H.S} = \bigcap_{i \in I} (x \in A_i \setminus B_i)$$

$$= \bigcap_{i \in I} (x \in A_i \wedge x \notin B_i)$$

$$= \bigcap_{i \in I} (x \in A_i) \wedge \bigcap_{i \in I} (x \notin B_i)$$

$$= \bigcap_{i \in I} A_i \wedge \neg \bigcup_{i \in I} B_i \quad \text{Double Negation}$$

$$= (\bigcap_{i \in I} A_i) \setminus (\bigcup_{i \in I} B_i) = \text{R.H.S}$$

$$14) (a) B_3 = \bigcup_{i \in I} A_{i,3}$$

$$= A_{1,3} \cup A_{2,3}$$

$$= \{1, 3, 4\} \cup \{2, 3, 5\}$$

$$= \{1, 2, 3, 4, 5\}$$

$$B_4 = \bigcup_{i \in I} A_{i,4}$$

$$= A_{1,4} \cup A_{2,4}$$

$$= \{1, 4, 5\} \cup \{2, 4, 6\}$$

$$= \{1, 2, 4, 5, 6\}$$

$$(b) \bigcap_{j \in J} B_j = B_3 \cap B_4$$

$$= \{1, 2, 4, 5\}$$

$$(c) \bigcup_{i \in I} (\bigcap_{j \in J} A_{i,j}) \quad (\because \text{think of it as loops})$$

$$= (A_{1,3} \cap A_{1,4}) \cup (A_{2,3} \cap A_{2,4})$$

$$= (\{1, 3, 4\} \cap \{1, 4, 5\}) \cup (\{2, 3, 5\} \cap \{2, 4, 6\})$$

$$= \{1, 4\} \cup \{2\}$$

$$= \{1, 2, 4\}$$

Not equal to set in (b)

$$(d) x \in \bigcap_{j \in J} (\bigcup_{i \in I} A_{i,j})$$

$$= \forall j \in J (x \in \bigcup_{i \in I} A_{i,j})$$

$$= \forall j \in J (\exists i \in I (x \in A_{i,j}))$$

$$= \forall j \in J \exists i \in I (x \in A_{i,j})$$

$$x \in \bigcup_{j \in J} (\bigcap_{i \in I} A_{i,j})$$

$$= \exists i \in I (x \in \bigcap_{j \in J} A_{i,j})$$

$$= \exists i \in I (\forall j \in J (x \in A_{i,j}))$$

$$= \exists i \in I \forall j \in J (x \in A_{i,j})$$

These 2 are not equivalent

15) $x \in \bigcup F = \exists A \in F (x \in A) = \exists A (A \in F \wedge x \in A)$

if $F = \emptyset$, then $\exists A \in F$ will be false, there is no such A , so statement is F for every x .

$\bigcup \emptyset$: Union of an empty family, since there is no set to take union, its result will be empty set

16) $x \in \bigcap F = \forall A \in F (x \in A) = \forall A (A \in F \rightarrow x \in A)$

$A \in F$ is F as F is $\emptyset$. In $p \rightarrow q$, if p is F, then $p \rightarrow q$ is T. so this statement is true, no matter what x is

$\forall A \in U$

16) (a) $(A \in R \rightarrow A \notin A)$

What about R ?

$(R \in R \leftrightarrow R \notin R)$ Contradiction

(b) Basically there can be no set that can be element of itself. Suppose $U = \{1\}$ then if U contains itself then $U = \{\{1\}\}$, But now U is different, so to contain itself $U = \{\{\{1\}\}\}$... $U \in U \in U$... infinite self reference (Stacking Problem).